Mitigate Catastrophic Remembering via Continual Knowledge Purification for Noisy Lifelong Person Re-Identification

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Abstract

Current Lifelong Person Re-Identification (LReID) methods focus on tackling a clean data stream with accurate labels. When noisy data with incorrect labels are given, their performance is severely degraded since the model inevitably and continually remembers erroneous knowledge induced by the label noises. Moreover, the well-known issue of catastrophic forgetting in LReID is exacerbated by noisy labels, which disrupt the retention of correct knowledge from previous models. Such a practical noisy LReID task is important but challenging, and rare works have attempted to handle it. In this paper, we initially investigate noisy LReID and propose a Continual Knowledge Purification (CKP) method to address the catastrophic remembering of erroneous knowledge and catastrophic forgetting of correct knowledge simultaneously. Specifically, a Cluster-aware Data Purification module (CDP) is designed to select clean labels based on clustering-guided label confidence estimation. Besides, an Iterative Label Rectification (ILR) pipeline is proposed to rectify wrong labels by fusing the prediction and label information throughout the training epochs. To handle the catastrophic remembering problem, an Erroneous Knowledge Filtering (EKF) algorithm is proposed to estimate and transfer the correct old knowledge to the new model. Finally, a Noisy LReID benchmark is constructed for performance evaluation and extensive experimental results demonstrate that our proposed CKP method achieves state-of-the-art performance. Our code is available at https://github.com/zhoujiahuan1991/MM2024-CKP

CCS Concepts

• **Computing methodologies** → **Object identification**.

Keywords

Lifelong Person Re-Identification, Label Noise Learning

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1 Introduction

Person re-identification (ReID) [1, 21] is a classical multimedia task that has been thoroughly investigated in stationary scenes [5, 10, 49]. Recently, lifelong person re-identification (LReID) [39], which aims to learn from the non-stationary data stream, has drawn increasing research attention. However, existing LReID methods assume that the training data is accurately annotated [26, 33]. In realistic scenarios, training labels are often inevitably noisy due to inaccurate person detection or annotation errors [4, 47], which has been shown to severely hinder stationary ReID performance [44, 52].

The LReID models are even more vulnerable to label noise which refers to the wrong labels shown in Figure 1 (a). As shown in Figure 1 (b), when noisy data are given for learning, the LReID model learns from erroneous identity-affinity supervision [45], leading to significant performance degradation. Besides, even though the model has acquired correct knowledge from historical data in Figure 1 (c), label noises in the new data can introduce erroneous information, overwriting the correct knowledge with erroneous knowledge [31, 56], thereby exacerbating catastrophic forgetting [54]. Additionally, as illustrated in Figure 1 (d), since the old model inevitably remembers erroneous knowledge of historical noisy data, the anti-forgetting strategy can cause erroneous knowledge accumulation and impede the acquisition of correct knowledge. These issues can significantly hinder the performance of the existing LReID methods [42]. Note that the histogram in Figure 1 (b) displays the performance when learning a dataset with clean and noise labels, respectively. The curves in Figure 1 (c) depict the performance trend on the first dataset, where the model learns with clean labels initially and subsequently adapts to datasets with varying noise ratios. Figure 1 (d) shows the average performance on all learned datasets with

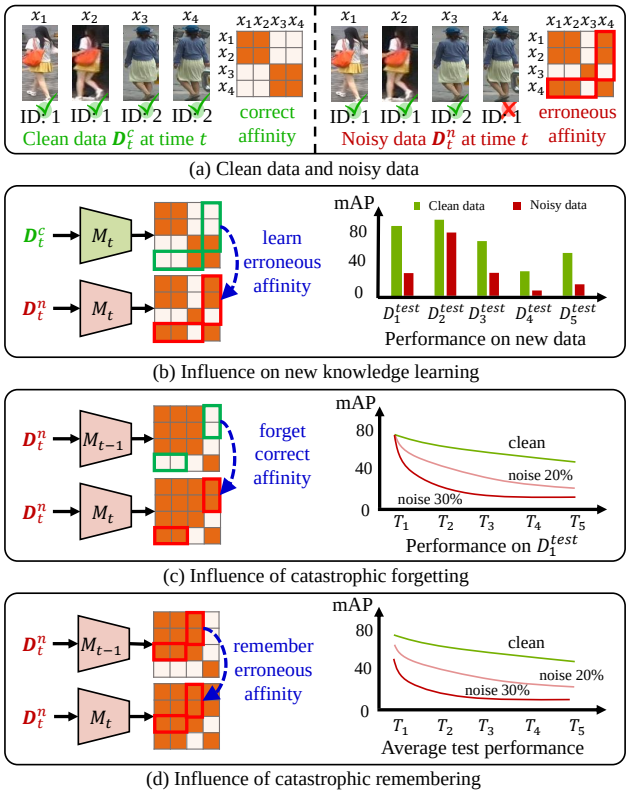


Figure 1: (a) Noisy data contains wrong labels which introduce erroneous inter-instance affinity knowledge. During LReID, noisy data not only (b) influences the knowledge learning of new datasets but also (c) exacerbates the forgetting of correct historical knowledge. Besides, (d) the learned erroneous knowledge is remembered by the new model M_t which severely hinders the LReID performance. The experiment results are obtained from the latest LReID method [42].

different noise ratios. The above results verify that label noise is a crucial and challenging problem for LReID.

Recently, several label noise learning (LNL) techniques have been explored to settle the noisy ReID and classification tasks [12, 47, 48]. They primarily rely on identity prediction results [47] or loss regularization strategies [48, 50] to mitigate the influence of noisy data [22, 35, 44]. However, these approaches neglect the disparity between label correctness and prediction/loss value distribution [36], *e.g.*, noisy samples near the distribution boundary tend to exhibit prediction scores or regularization effects that are indistinguishable from clean samples. Thus, some wrongly labeled samples can easily be confused with clean ones, leading to the acquisition of erroneous knowledge. Therefore, when directly applying these LNL methods to Noisy LReID, the issues illustrated in Figure 1 (b), (c), and (d) remain critical.

In this paper, we initially investigate this challenging noisy LReID task and propose a novel method named Continual Knowledge Purification (CKP) to handle the catastrophic remembering and forgetting issues. Specifically, a Cluster-aware Data Purification module (CDP) and an Iterative Label Rectification pipeline (ILR) are proposed in our CKP to achieve high-quality clean data. CDP can adaptively select the clean samples for new model training by estimating the label confidence based on feature clustering, thereby settling the problems in Figure 1 (b) and (c). Instead of simply discarding the wrongly labeled samples, the proposed ILR aims to fully utilize them by rectifying their labels along with model learning, so that these data can be recollected by CDP for reuse. Furthermore, to actively forget the erroneous old knowledge, an Erroneous Knowledge Filtering algorithm (EKF) is proposed to estimate the knowledge correctness of the old model outputs, and a weighted knowledge distillation loss is designed to transfer the correct old knowledge to the new model while excluding the erroneous one. Thus, the erroneous knowledge remembering issue in Figure 1 (d) could be greatly mitigated. To evaluate the performance of our method, a Noisy LReID Benchmark (NLReID) is proposed which is inspired by the existing LReID and LNL benchmark configurations [26, 47]. Extensive experimental results under various noisy conditions demonstrate the superiority of our CKP model.

In summary, the contributions of this work are three-fold: (1) We provide a pioneer investigation on the important and challenging Noisy LReID task, and a comprehensive Noisy LReID benchmark (NLReID) is proposed for the evaluation of different methods. (2) To handle the catastrophic remembering and forgetting issues, a novel Continual Knowledge Purification (CKP) method is proposed. A Cluster-aware Data Purification module and an Iterative Label Rectification pipeline are designed to obtain cleaner training data for correct new knowledge learning and mitigating erroneous new knowledge acquisition. Besides, an Erroneous Knowledge Filtering algorithm is developed to actively forget erroneous old knowledge and ensure correct new knowledge remembering. (3) Extensive experiments demonstrate that our CKP achieves state-of-the-art Noisy LReID performance, and the proposed method can be readily integrated with the latest LReID or LNL approaches to further improve the performance in the noisy LReID scenario.

2 Related Work

2.1 Lifelong Person Re-Identification

Lifelong person re-identification (LReID) [26, 39] aims to train a ReID model with non-stationary data, improving the model's adaptability to various conditions. Existing LReID works [6, 26–28, 33, 39, 51] primarily focus on alleviating the catastrophic forgetting problem, which indicates that the performance of the model on historical data is degraded greatly when the new data is learned [13, 34, 37, 46]. Nevertheless, these methods assume that the training data are all correctly annotated [26, 39]. However, in real scenarios, the training data labels are often noisy due to inaccurate person detection or annotation errors [3, 4, 47]. Such a Lifelong Person Re-Identification with the Noisy Label (Noisy LReID) scenario is more challenging since not only the correct knowledge catastrophic forgetting exacerbated due to erroneous new knowledge continually overwriting the correct old knowledge, but also catastrophic remembering [15, 53] issues occur as the erroneous knowledge from different domains accumulates, resulting in degraded performance on new domains.

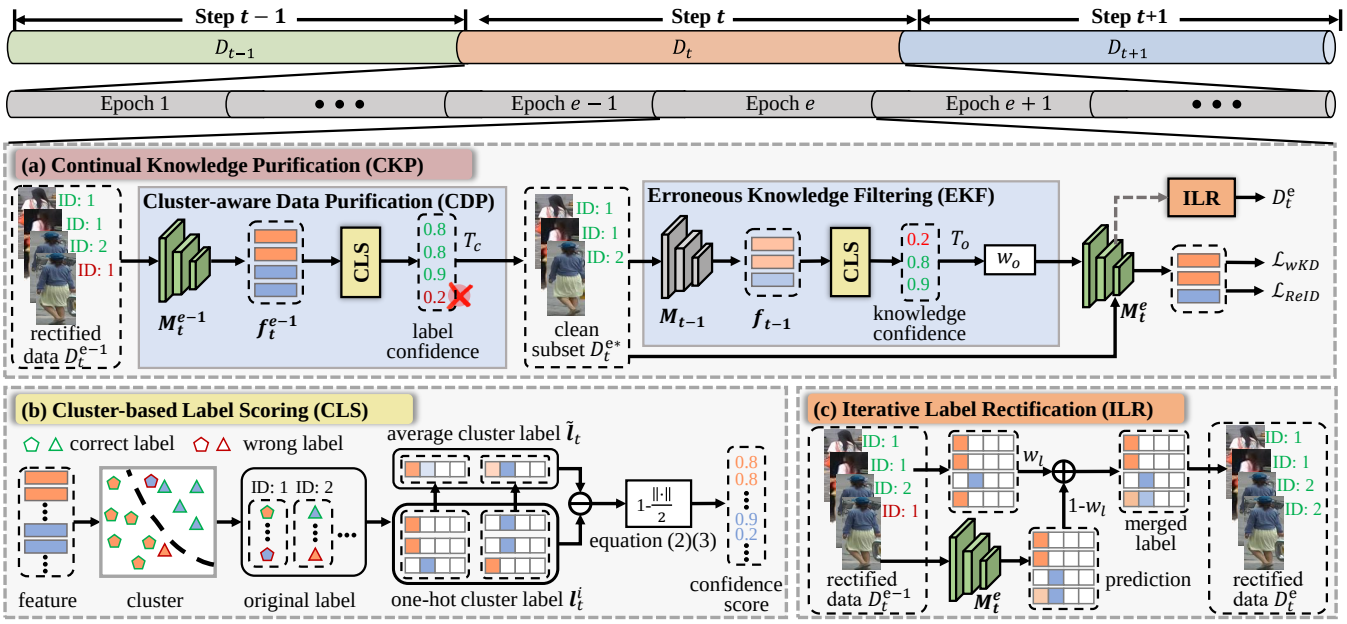


Figure 2: Given the noisy dataset D_t at the t-th LReID step, our CKP updates the rectified dataset D_t^e and model M_t^e along the training epoch e. CDP aims to obtain a clean subset D_t^{e*} for M_t^e learning. Besides, EKF aims to filter the features containing erroneous old knowledge. Finally, ILR rectifies the noise labels by fusing the learning knowledge and label information.

2.2 Label Noise Learning

Label Noise Learning (LNL) has been widely investigated in recent years [17, 32, 57]. Most existing LNL methods rely on identity prediction [12, 44, 47] or loss regularization strategies [22, 48, 50] to filter noisy data or accomplish noise-robust learning. For example, LCNL [45] adopts Gaussian Mixture Model (GMM) [35, 44] to model the loss distribution and select the unreliable samples. However, existing works reveal that there is a discrepancy between the prediction score/loss and the actual identity distribution [36]. Specifically, the noisy samples around the distribution boundary exhibit similar prediction scores or regularization effects compared to the clean ones. Thus, these samples can easily be confused with clean labels. , resulting in inaccurate filtering or invalid regularization. Therefore, the learned models still contain considerable erroneous knowledge, and the catastrophic remembering problem is still critical.

2.3 Lifelong Learning with Label Noise

Lifelong Learning with Label Noise problem has yet to gain widespread attention and existing solutions focus on classification task [9, 14, 16]. These methods rely on filtering and retaining historical exemplars to address the catastrophic forgetting issue during lifelong learning. However, as human images are highly privacy-sensitive data, retaining historical exemplars is not feasible in many actual applications [26, 33]. Therefore, in this paper, we provide a pioneer investigation of the Noisy LReID problem, the catastrophic forgetting of correct knowledge and the catastrophic remembering of erroneous knowledge under such a scenario is thoroughly discussed, and a novel exemplar-free Noisy LReID method is proposed.

3 Continual Knowledge Purification for Noisy LReID

3.1 Problem Definition and Formulation

Noisy lifelong person re-identification (Noisy LReID) aims to continually learn from a stream of T ReID datasets $\mathcal{D}^{tr} = \{D_t\}_{t=1}^{T}$, each containing a certain ratio of noisy labels. The effectiveness of the final model is evaluated on the clean test sets $\mathcal{D}^{te} = \{D_t^{te}\}_{t=1}^{T}$ corresponding to each domain, to evaluate the new knowledge acquisition and anti-forgetting capacity of the model. Besides, a series of additional U clean test sets $\mathcal{D}^{un} = \{D_t^{un}\}_{t=1}^{U}$ are tested to evaluate the generalization of the models on unseen domains. In this paper, the model learned after training step t is denoted as M_t and the intermediate model after each training epoch e is denoted as M_t^e. The parameters of M_t^0 are initialized with M_{t-1}.

3.2 Overview

As is shown in Figure 2, given the noisy training dataset D_t at training step t, our overall approach generates a rectified dataset D_t^e at the training epoch e, where D_t^0 is initialized with original noisy data D_t. The proposed Continual Knowledge Purification (CKP) framework consists of three key components, *i.e.*, Cluster-aware Data Purification (CDP), Erroneous Knowledge Filtering (EKF), and Iterative Label Rectification (ILR). Specifically, the CDP module aims to estimate label confidence for each instance and generate a clean subset D_t^{e*} to ensure accurate new data learning. Then, EKF is employed to estimate the knowledge correctness of the old model features, so that the erroneous knowledge could be actively forgotten and the correct new knowledge could be consolidated.

Besides, at the end of the e-th epoch, the IRL pipeline is adopted to rectify the noisy labels by fusing the model prediction and label information, resulting in the rectified dataset D_t^{e+1} for subsequent epochs. Since the label confidence estimation function serves as a crucial component for CDP and EKF modules, we introduce the proposed Cluster-aware Label Scoring (CLS) strategy first and then depict the designs of CDP, EKF, and ILR sequentially.

3.3 Cluster-aware Label Scoring

In this work, we propose to utilize clustering [43] technology to gather instances with shared characteristics to achieve reliable label confidence estimation. Specifically, given the noisy dataset $\{(x_i, y_i)\}_{i=1}^{N^t}$ with N_t images x_i and corresponding labels y_i, the extracted features are $\{f_i\}_{i=1}^{N_t}$. The DBSCAN algorithm [30] is adopted to generate clusters with different shared characteristics and each instance is assigned a cluster-aware label $\tilde{y}_i \in \{1, 2, ..., N_c\}$ where N_c is the cluster number. Note that the outliers during the clustering process are collected as an extra cluster whose label is set to N_c. Then, we generate one-hot embedding $\boldsymbol{l}_i \in \mathbb{R}^{N_c}$ for all instances.

Then, to bridge the connection between the annotated label and the generated cluster label, an annotation-aware average cluster label $\bar{\boldsymbol{l}}_i$ for each instance x_i is calculated by

$$\bar{\boldsymbol{l}}_i = \frac{1}{n_t^i} \sum_{j=1}^{N_t} \delta(y_i, y_j) \boldsymbol{l}_j, \tag{1}$$

where $n_t^i = \sum_{j=1}^{N_t} \delta(y_i, y_j)$, and $\delta(y_i, y_j)$ is a sign function that outputs 1 and 0 when $y_i = y_j$ and $y_i \neq y_j$ respectively. Equation (1) indicates that given x_i with annotated label y_i, obtain the average cluster label of all instances x_j that possess the same annotated label as y_i. Therefore, $\bar{\boldsymbol{l}}_i$ is consistent across instances with the same annotated label, representing the aggregate value of each annotated label in the cluster label space.

To quantify the label confidence of each instance x_i, the label distance d_i is defined as the squared L2 norm between l_i and $\bar{\boldsymbol{l}}_i$, represented as:

$$d_i = ||\boldsymbol{l}_i - \bar{\boldsymbol{l}}_i||_2^2, \tag{2}$$

where d_i measures the disparity between the cluster label $\tilde{y}_i$ and the annotated identity centers. Note that $d_i \in [0, 2)$ and the d_i values of outliers during clustering are adjusted to 2. This adjustment is made because the outliers exhibit minimal resemblance to other instances and thus possess the lowest confidence.

Then the annotated label confidence score s_i is calculated by

$$s_i = (2 - d_i)/2, \tag{3}$$

where $s_i \in [0, 1]$ with higher values indicating greater trustworthiness of the annotated label.

Discussion: Existing label confidence scoring methods primarily utilize the Gaussian Mixture Model (GMM) to model the noise distribution for clean data selection [12, 45]. However, since there is a discrepancy between the model loss and the identity correctness distribution [36], GMM can reserve many noisy samples, thereby limiting the new knowledge acquisition and correct knowledge anti-forgetting capacity (Figure 1 (a)(b)). However, the elaborately designed CLS strategy can effectively mine fine-grained inter-instance similarity to evaluate identity coherence across instances, therefore the intra-identity distribution is fully modeled and utilized to enhance the reliability of label confidence estimation results.

3.4 Cluster-aware Data Purification

In this paper, given the noisy dataset D_t, we iteratively update the labels of the images as introduced in Section 3.6. The updated dataset after e-th epoch is denoted as D_t^e. As is shown in Figure 2, at the e-th epoch, D_t^{e-1} and M_t^{e-1} is used to process the images and obtain the features $\mathcal{F}_t^e = \{f_t^i\}_{i=1}^{N_t}$. Then the CLS strategy is adopted to process $\mathcal{F}_t^e$ to get the label confidence s_i of each image x_i.

Then, we remove the data of low-confidence labels with a confidence threshold T_c and obtain the clean subset D_t^{e*}:

$$D_t^{e*} = \{(x_i, y_i)\}_{i=1}^{N_t^*}, \tag{4}$$

where each instance x_i in D_t^{e*} has the s_i higher than T_c and N_t^* is the selected instance number.

3.5 Erroneous Knowledge Filtering

The knowledge distillation [7] strategy is a widely-used anti-forgetting approach adopted by the existing LReID methods [6, 26, 33, 39]. Despite its knowledge consolidation capacity, such a strategy can lead to erroneous knowledge accumulation and even mislead the learning of the new data, as shown in Figure 1 (d).

Therefore, in this section, we aim to filter the features of samples that reflect the learned correct knowledge of old model M_{t-1} and discard the features that contain erroneous knowledge contained in M_{t-1}. Specifically, given the clean subset D_t^{e*}, we utilize M_{t-1} to process the D_t^{e*} and the generated features $\{f_{t-1}^i\}_{i=1}^{N_t^*}$ are fed into the CLS (Section 3.3), where the obtained scores of each instance is named feature confidence s_i^o. Then a knowledge distillation weight w_i^o is assigned to each instance x_i by

$$w_i^o = \begin{cases} 0 & s_i^o \leq T_o \\ 1 & s_i^o > T_o \end{cases}, \tag{5}$$

where T_o is the hyperparameter serving as the threshold of s_i^o. Notably, $w_i^o = 0$ indicates that the old knowledge can not correctly process the instance x_i, thus f_{t-1}^i primarily contains erroneous old knowledge and should be discarded during knowledge distillation.

Then we proposed a weighted knowledge distillation loss $\mathcal{L}_{wKD}$ to ensure correct old knowledge transfer and active erroneous old knowledge forgetting. Considering there are primarily two kinds of knowledge distillation loss, *i.e.* logits-based and inter-instance relation-based. We design the $\mathcal{L}_{wKD}$ variants accordingly.

Specifically, as for logits-based knowledge distillation loss [23, 26], w_i^o serve as the weight of each instance directly:

$$\mathcal{L}_{wKD-lgs} = w_i^o \mathcal{L}_{KD}(M_{t-1}(x_i), M_t^e(x_i)), \tag{6}$$

where $\mathcal{L}_{KD}$ is a ordinary loss function, *e.g.*, KL-diverigence [42], MSE [25]. As for the inter-instance relation-based knowledge distillation, given a batch of instances $\mathcal{B}$, a maximum subset $\mathcal{B}_o$ where each instance with $w_i^o = 1$ is selected to calculate the inter-instance relation loss:

$$\mathcal{L}_{wKD-rel} = \mathcal{L}_{KD}\left(\theta_r\left(M_{t-1}\left(\mathcal{B}_o\right), M_t\left(\mathcal{B}_o\right)\right)\right), \tag{7}$$

Table 1: Results under the Random Noise. † indicates the state-of-the-art LNL method is combined with the latest anti-forgetting strategy of LSTKC.

Metric	Type	Method	Market-1501			CUHK-SYSU			DukeMTMC			MSMT17			CUHK03			Seen-Avg			Unseen-Avg		
			10%	20%	30%	10%	20%	30%	10%	20%	30%	10%	20%	30%	10%	20%	30%	10%	20%	30%	10%	20%	30%
mAP	LReID	LwF [23]	49.2	29.1	19.2	66.2	53.7	47.7	19.0	9.8	7.5	3.6	1.9	1.3	17.3	8.9	6.1	31.1	20.7	16.4	35.4	25.6	20.2
		PatchKD [33]	**51.0**	30.6	23.9	66.8	58.0	52.6	19.3	11.9	8.4	3.6	2.1	1.5	18.1	10.3	6.4	31.8	22.6	18.6	35.0	27.3	23.8
		KRKC [51]	29.4	20.1	14.8	71.7	65.3	59.2	23.3	15.3	10.3	5.1	3.1	2.2	35.3	20.9	12.7	33.0	24.9	19.8	42.5	35.4	29.0
		DKP [41]	43.2	28.1	20.4	78.4	71.0	66.7	34.8	22.5	15.4	12.8	7.7	5.5	21.2	10.7	6.6	38.1	28.0	22.9	47.4	37.6	32.5
		LSTKC [42]	41.4	35.1	25.8	78.7	74.2	67.4	39.7	19.4	9.3	14.8	5.2	2.6	27.8	15.3	6.9	40.5	29.8	22.4	47.2	37.0	30.0
	LNL	CORE [47]	35.5	29.4	23.7	75.7	73.7	69.2	36.3	30.4	22.1	12.5	9.9	6.5	41.8	35.2	24.9	40.4	35.7	29.3	51.1	47.3	40.4
		DICS [22]	34.0	23.1	15.7	72.1	70.1	63.6	34.8	22.3	13.1	12.2	7.3	4.5	35.5	18.5	9.3	37.7	28.3	21.2	48.0	38.8	32.2
	Noisy LReID	DICS† [22]	38.3	38.3	29.4	73.4	73.4	67.4	37.1	37.1	14.0	8.8	8.8	3.2	13.1	13.1	6.4	34.1	34.1	24.1	41.5	41.5	32.7
		CORE† [47]	48.9	**45.2**	37.8	**81.6**	80.2	74.7	46.2	37.1	21.3	**18.2**	10.5	5.5	35.5	23.9	15.3	46.1	39.4	30.9	52.9	47.2	40.7
		LCNL†[45]	28.9	23.5	16.2	69.2	67.9	63.1	29.4	20.0	11.6	9.7	6.8	4.3	38.1	27.8	18.6	35.1	29.2	22.8	45.1	38.4	33.8
		CKP (Ours)	48.7	44.5	**42.2**	80.8	**80.3**	**78.6**	**47.3**	**44.4**	**42.1**	18.1	**16.4**	**14.6**	**42.0**	**36.3**	**33.5**	**47.4**	**44.4**	**42.2**	**56.0**	**51.4**	**50.4**
R@1	LReID	LwF [23]	**74.2**	55.9	43.0	69.9	57.4	51.4	35.5	20.3	17.9	11.3	6.7	5.0	17.4	8.6	5.5	41.7	29.8	24.6	28.9	20.5	15.5
		PatchKD [33]	**74.2**	56.9	49.5	70.5	61.7	56.0	34.1	23.1	18.1	11.0	7.2	5.5	17.2	9.8	5.9	41.4	31.7	27.0	29.6	21.7	18.9
		KRKC [51]	54.0	42.0	34.4	75.1	68.8	63.4	38.4	28.0	19.8	14.9	9.8	7.5	37.1	19.6	11.1	43.9	33.6	27.2	36.9	29.4	24.3
		DKP [41]	68.7	54.3	45.7	81.1	74.8	70.4	53.9	39.5	30.4	32.5	23.3	18.2	20.4	9.9	5.5	51.3	40.4	34.0	41.0	32.0	26.8
		LSTKC [42]	66.8	60.5	50.2	81.3	77.6	70.9	59.3	34.0	19.5	35.5	16.2	8.8	27.7	14.6	6.2	54.1	40.6	31.1	40.3	30.7	25.0
	LNL	CORE [47]	63.0	55.8	49.8	78.7	77.0	72.6	56.7	48.9	38.8	32.9	26.9	19.0	**43.1**	36.2	24.1	54.9	49.0	40.9	44.7	41.7	34.5
		DICS [22]	58.6	48.5	38.2	74.9	74.2	67.9	55.7	40.6	27.2	33.1	23.0	16.6	36.3	18.3	8.6	51.7	40.9	31.7	42.0	33.4	27.3
	Noisy LReID	DICS† [22]	65.6	65.6	56.4	76.8	76.8	70.9	57.3	57.3	28.1	26.7	26.7	12.4	12.8	12.8	6.2	47.8	47.8	34.8	34.9	34.9	27.3
		CORE† [47]	72.6	**69.9**	62.8	**83.9**	**83.0**	77.4	63.8	55.5	37.5	**41.0**	27.1	16.8	36.4	22.9	14.7	59.5	51.7	41.8	46.0	40.9	35.3
		LCNL†[45]	55.3	48.8	37.6	72.7	71.8	66.9	49.1	36.3	23.1	27.6	21.0	14.7	39.7	29.4	18.9	48.9	41.5	32.2	38.4	32.6	28.5
		CKP (Ours)	71.8	68.1	**66.9**	83.2	**83.0**	**81.0**	**64.7**	**62.1**	**58.9**	40.1	**37.5**	**34.7**	42.4	**37.1**	**34.1**	**60.4**	**57.6**	**55.1**	**49.5**	**44.6**	**43.4**

where θ_r is a relation evaluation function [42] and $\mathcal{L}_{KD}$ is a relation knowledge distillation loss [33].

3.6 Iterative Label Rectification

Although the above CDP and EKF modules could ensure the model learning the correct knowledge, the wrongly labeled data which can contain abundant information are discarded. To settle this drawback, we propose to rectify the annotated data iteratively along the model learning epochs, ensuring the correct knowledge learning and enhancing informative data utilization simultaneously.

Specifically, the label rectification is accomplished by

$$y_i^* = \text{argmax}\{\boldsymbol{y}_i * w_l + \hat{\boldsymbol{y}}_i * (1 - w_l)\}, \tag{8}$$

where $\boldsymbol{y}_i \in \mathbb{R}^{N_p}$ is a one-hot embedding generated from annotated label y_i in D_t^e and N_p is the annotated person identity number. The label rectification weight w_l aims to fuse the information of annotation and prediction. $\hat{\boldsymbol{y}}_i \in \mathbb{R}^{N_p}$ is the identity prediction vector generated by M_t^e. y_i^* is utilized to replace y_i in D_t^e to obtain new dataset D_t^{e+1}. Note that ILR is not necessarily conducted after each epoch, and a rectification interval of e_0 epochs is adopted for computational efficiency.

Model Training During training, our framework can be integrated with existing LReID and LNL methods by introducing their noisy data learning loss $\mathcal{L}_{ReID}$ and our weighted knowledge distillation loss $\mathcal{L}_{wKD}$ (Equation (6) and (7)). Therefore, the overall loss is calculated by:

$$\mathcal{L} = \mathcal{L}_{ReID} + \mathcal{L}_{wKD}. \tag{9}$$

Model Inference We follow existing methods [42] to use the feature generated by the final model M_T for person matching.

4 Experiments

4.1 Benchmark

In this paper, following the existing LReID [26] and Noisy ReID [47] works, a Noisy LReID benchmark (NLReID) is proposed as below.

Datasets: NLReID contains 12 ReID datasets, 5 of them are used for lifelong training and evaluation (Market1501 [58], DukeMTMC-reID [29], CUHK-SYSU [40], MSMT17-V2 [38], and CUHK03 [20]), and the other 7 test datasets are used evaluate the generalizability of the model (CUHK01 [19], CUHK02 [18], VIPeR [8], PRID [11], i-LIDS [2], GRID [24], and SenseReID [55]). The clean identity configuration of each dataset follows the setting of LReID [26].

Label Noise Generation: Two different noisy settings are considered [47]. (1) *Random Noise* means a certain percentage (10%, 20%, 30%) of training data are assigned with random labels of other identities. (2) *Patterned Noise* means a certain percentage (10%, 20%,

Table 2: Results under the Patterned Noise. † indicates the state-of-the-art LNL method is combined with the latest anti-forgetting strategy of LSTKC.

Metric	Type	Method	Market-1501			CUHK-SYSU			DukeMTMC			MSMT17			CUHK03			Seen-Avg			Unseen-Avg		
			10%	20%	30%	10%	20%	30%	10%	20%	30%	10%	20%	30%	10%	20%	30%	10%	20%	30%	10%	20%	30%
mAP	LReID	LwF [23]	47.8	33.2	17.4	60.8	47.3	32.5	22.6	11.6	5.9	4.4	2.0	1.0	11.9	5.7	3.3	29.5	20.0	12.0	35.4	23.8	19.0
		PatchKD [33]	**50.6**	34.2	18.9	62.1	48.2	34.7	22.1	12.1	5.9	4.5	2.2	1.1	11.4	5.5	2.9	30.1	20.4	12.7	35.1	25.5	18.5
		KRKC [51]	31.9	22.3	18.0	73.1	66.7	63.5	27.1	16.2	10.4	5.7	3.4	2.6	39.1	24.8	16.1	35.4	26.7	22.1	45.4	36.9	31.9
		DKP [41]	46.1	33.7	27.4	80.3	73.8	70.1	37.4	26.1	20.3	14.2	9.2	7.2	24.8	14.9	9.3	40.6	31.5	26.9	50.5	41.2	36.7
		LSTKC [42]	40.3	36.0	33.2	78.3	75.4	71.6	38.3	30.6	18.0	15.2	8.0	4.4	30.2	20.0	13.2	40.5	34.0	28.1	50.8	42.2	37.9
	LNL	CORE [47]	35.7	33.4	30.1	75.8	74.8	73.3	36.9	30.9	29.2	13.5	11.1	9.3	**42.4**	37.7	31.5	40.9	37.6	34.7	52.2	46.5	45.0
		DICS [22]	32.3	26.1	19.1	69.3	71.3	67.3	34.2	26.6	16.1	10.8	9.5	5.7	35.8	23.6	13.2	36.5	31.4	24.3	45.9	43.7	35.8
	Noisy LReID	DICS† [22]	40.5	39.5	31.4	77.4	75.0	68.1	40.1	39.5	33.8	16.4	9.0	6.9	33.7	21.1	16.4	41.6	36.8	31.3	52.7	47.4	41.1
		CORE† [47]	49.0	**49.6**	**45.4**	80.8	**80.0**	**79.0**	46.4	42.2	38.7	**19.2**	9.2	10.8	39.5	32.2	21.7	47.0	42.6	39.1	55.8	48.6	47.6
		LCNL†[45]	29.6	25.9	21.1	68.4	67.6	65.6	29.2	23.0	19.6	9.7	7.6	6.2	38.5	31.5	23.6	35.1	31.1	27.2	45.1	40.7	38.0
		CKP (Ours)	50.1	46.9	44.1	**81.0**	79.9	78.9	**47.2**	**45.2**	**43.1**	18.3	**17.1**	**15.7**	41.9	**39.9**	**36.6**	**47.7**	**45.8**	**43.7**	**57.3**	**54.4**	**51.1**
R@1	LReID	LwF [23]	72.0	59.8	37.5	63.8	49.3	32.7	42.1	25.0	14.5	14.5	7.0	4.4	12.8	6.7	3.8	41.0	29.6	18.6	28.3	17.8	13.7
		PatchKD [33]	**74.6**	60.2	40.8	64.7	50.9	36.4	40.8	26.8	15.7	14.2	7.7	4.6	11.6	6.1	3.2	41.2	30.3	20.1	28.9	20.1	13.1
		KRKC [51]	57.5	44.7	38.5	76.5	70.4	67.0	43.4	28.6	20.0	16.2	10.9	8.6	40.3	23.9	15.9	46.8	35.7	30.0	38.5	32.2	27.0
		DKP [41]	71.1	60.2	53.8	82.9	77.2	74.1	55.4	42.8	36.8	34.8	25.9	21.5	24.8	14.3	7.6	53.8	44.1	38.8	43.4	35.9	30.7
		LSTKC [42]	65.2	59.9	58.1	80.8	78.0	74.8	56.3	48.6	32.4	36.8	22.5	13.6	31.3	20.5	12.9	54.1	45.9	38.4	44.3	36.1	32.2
	LNL	CORE [47]	63.4	60.1	57.1	78.7	77.9	76.7	57.3	50.7	46.7	34.1	29.6	25.6	**43.8**	38.2	32.4	55.5	51.3	47.7	45.9	40.1	38.9
		DICS [22]	57.3	51.9	43.2	72.5	75.0	71.3	55.1	44.5	31.2	29.7	28.4	19.3	36.4	24.1	13.0	50.2	44.8	35.6	38.7	37.3	30.3
	Noisy LReID	DICS† [22]	65.7	64.5	58.9	79.5	78.0	71.9	58.8	59.2	54.8	40.1	26.8	22.9	35.1	21.6	16.4	55.8	50.0	45.0	46.5	40.9	35.0
		CORE† [47]	72.6	**72.8**	**69.9**	**82.8**	**82.3**	**81.6**	65.0	59.0	56.3	**43.4**	24.6	28.1	40.5	31.9	21.2	60.9	54.1	51.4	48.7	42.1	40.8
		LCNL†[45]	55.8	51.6	47.0	71.5	71.1	69.5	49.2	41.1	35.7	27.2	23.0	19.4	39.6	33.3	24.6	48.7	44.0	39.2	38.4	34.9	32.3
		CKP (Ours)	73.2	70.8	67.5	82.7	**82.3**	81.4	**65.8**	**62.5**	**61.2**	41.0	**38.9**	**36.7**	42.7	**40.8**	**37.0**	**61.1**	**59.1**	**56.8**	**50.1**	**47.5**	**44.6**

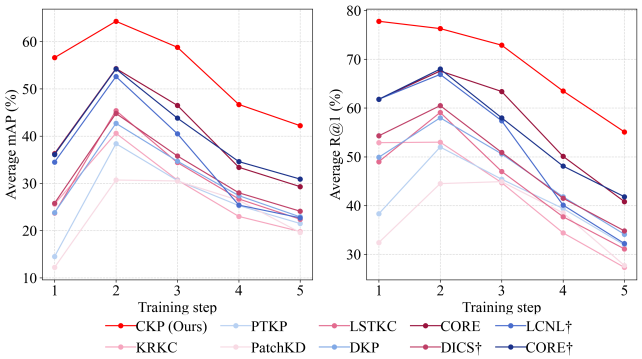


Figure 3: Seen domain knowledge consolidation capacity under 30% random noise.

30%) of training images are assigned with the labels of its most similar sample from other identities where the similarity is evaluated by a base model pre-trained with clean labels.

Evaluation Metrics: Following existing LReID works [6, 26, 33], the mean Average Precision (mAP) and Rank@1 accuracy (R@1) are adopted to evaluate the model performance on each seen and unseen dataset. Additionally, the seen/unseen average mAP and R@1 are reported to compare the lifelong learning and generalization capacity of the models across different scenarios.

4.2 Implementation Details

The state-of-the-art LReID method [42] is used as our baseline. During training, the first dataset is trained for 80 epochs and the subsequent datasets are trained for 60 epochs. 32 identities with 4 images for each identity are sampled as a mini-batch. The learning rate and weight decay are set as 0.008 and 0.0001 respectively, and an SGD optimizer is adopted. The hyperparameters T_c, T_o, w_l, and e_0 are set to 0.8, 0.2, 0.1, and 5 respectively.

4.3 The Comparison Methods

To comprehensively evaluate our method, extensive state-of-the-art LReID approaches (PatchKD [33], KRKC [51], LSTKC [42] and DKP [41]) are compared. The well-known class incremental learning method LwF [23] is also included. In addition, we combine the state-of-the-art LReID method, LSTKC, with the latest noisy label learning methods (LCNL [45], DICS [22], and CORE [47]) to adapt them to the Noisy LReID scenario. All the methods above are implemented with the official codes, and we ensure a fair comparison by adopting the same backbone and data configurations.

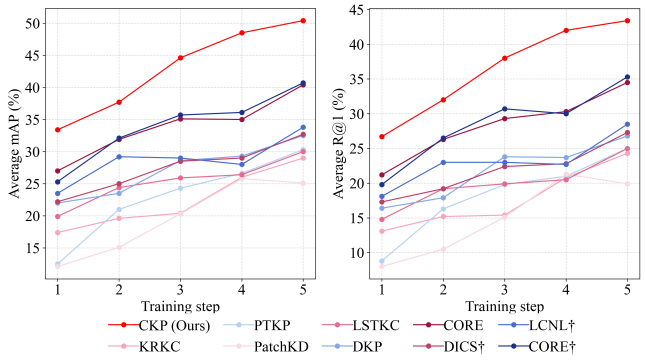


Figure 4: Unseen domain generalization capacity under 30% random noise.

4.4 Comparison with state-of-the-art methods

The results of different methods on the NLReID benchmark are reported in Table 1 and Table 2 under different ratios of random and patterned noise respectively. The best results under each scenario are highlighted in **Bold**.

Compared to LReID Methods: As shown in Table 1 and Table 2, our CKP achieves significantly superior performance on the average performance of both seen and unseen domains compared to LReID methods. Specifically, when the random noise ratio increases from 10% to 30%, these methods exhibit degradation in average mAP and R@1 performance ranging from **12%-21%** on both seen and unseen domains. In contrast, our model experiences no more than 6.1% degradation, attributed to the effectiveness of the new data purification and erroneous old knowledge filtering designs.

Compared to LNL Methods: In Table 1 and Table 2, our CKP outperforms state-of-the-art LNL method CORE significantly. Specifically, CKP achieves average mAP/R@1 improvement of **12.9%/14.2%** and **10.0%/8.9%** under seen and unseen domains when learning under 30% random noise, and **12.4%/9.1%** and **10.0%/5.7%** improvement when learning under 30% patterned noise is obtained. The results arise because LNL methods are designed for stationary scenarios, neglecting the catastrophic forgetting of correct knowledge and the catastrophic remembering of erroneous knowledge.

Compared to Noisy LReID Methods: We incorporate the anti-forgetting strategy of the latest LReID method LSTKC, into LNL methods, obtaining the Noisy LReID approaches DICS†, CORE†, and LCNL†. Among them, CORE† exhibits the highest average performance on different settings. As is shown in Table 1, compared to CORE† under random noise, we achieve the improvement of **1.3%/0.9%**, **5.0%/5.9%**, and **11.3%/13.3%** on average mAP/R@1 performance in seen domains under noise ratios of **10%**, **20%**, and **30%**, respectively. Additionally, we also obtain the improvement of **3.1%/3.5%**, **4.2%/3.7%**, and **9.7%/8.1%** in the average mAP/R@1 performance of unseen domains under noise ratios of **10%**, **20%**, and **30%**, separately. As is shown in Table 2, under the challenging patterned noise, our CKP consistently obtains **0.7%/0.2%**, **3.2%/5.0%**, and **4.6%/5.4%** improvement on the average mAP/R@1 of seen domains under **10%**, **20%**, and **30%** noise. The increasing improvement under higher noise ratios highlights the superiority of our method in mining correct knowledge and reducing the remembering of erroneous knowledge in noisy scenarios.

Table 3: Ablation study of different components in CKP under 30% random noise.

Baseline	CDP	ILR	EKF	Seen-Avg mAP	Seen-Avg R@1	Unseen-Avg mAP	Unseen-Avg R@1
✓				30.9	41.8	40.7	35.3
✓	✓			38.8	51.6	47.3	40.2
✓		✓		34.5	45.7	43.5	37.4
✓			✓	34.8	46.1	43.8	37.3
✓	✓	✓		41.9	54.7	49.2	43.1
✓	✓	✓	✓	**42.2**	**55.1**	**50.4**	**43.4**

Seen Domain Performance Curves. To show the new knowledge acquisition and anti-forgetting capacity of different models, We test the models on seen domains after each learning stage. The results are shown in Figure 3. Compared to the competitors, our method outperforms them in the first dataset and maintains superiority throughout the training process. These results show that our proposed method could consistently consolidate correct knowledge by learning from the noise data of various domains.

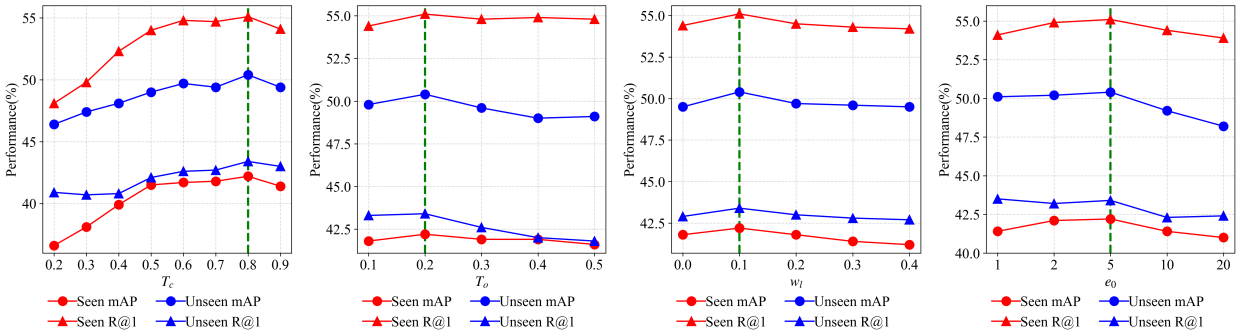
Unseen Domain Generalization Curves. We further visualize the average performance on the unseen domains along the lifelong training steps, as depicted in Figure 4. The results demonstrate that our proposed model outperforms existing methods in capturing more generalizable knowledge when learning from non-stationary noisy data. This result is attributed to the knowledge purification mechanism of our model that ensures correct knowledge mining and erroneous knowledge filtering.

4.5 Ablation Studies

In this section, we evaluate and discuss the effectiveness of our proposed components. All experiments are conducted on 30% random noise data in the NLReID benchmark.

Ablations on different components. In Table 3, we start with a CORE† baseline and progressively integrate the proposed CDP, ILR, and EKF modules. The results illustrate that each module improves the model performance when utilized independently, and their combined utilization further boosts performance. Particularly noteworthy is the significant improvement yielded by CDP, underscoring the critical importance of ensuring training data clarity in mitigating label noise impact.

Ablations on hyperparameters. We analyze the effects of the hyperparameters T_c, T_o, w_l, and e_0, on the model in Figure 5. The results in Figure 5 (a) show that a relatively high T_c helps improve the overall performance of the model, highlighting the importance of training data purity. In Figure 5 (b), we observe that an optimal T_o tends to be relatively small, as some features may contain both correct and erroneous knowledge simultaneously. And T_o=0.2 shows the best balance. The results in Figure 5(c) suggest that a relatively low w_l is optimal for label rectification, indicating that the model prediction is more reliable than the annotated label, yet the annotated label still contains some crucial clues that could remedy the imperfect predictions of the model. Figure 5 (d) shows that



(a) label confidence threshold T_c (b) feature confidence threshold T_o (c) label rectification weight w_l (d) label rectification interval e_0

Figure 5: Ablation studies on hyperparameters under 30% random noise. Dashed green lines highlight our default values.

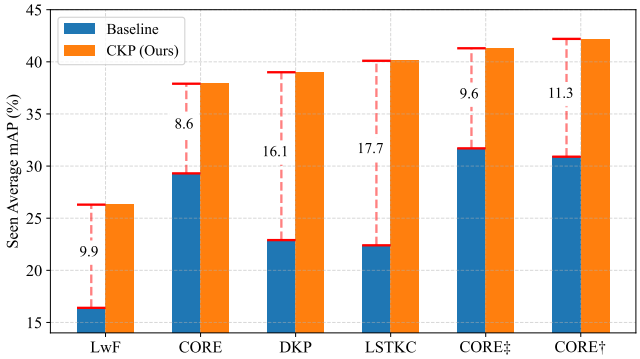
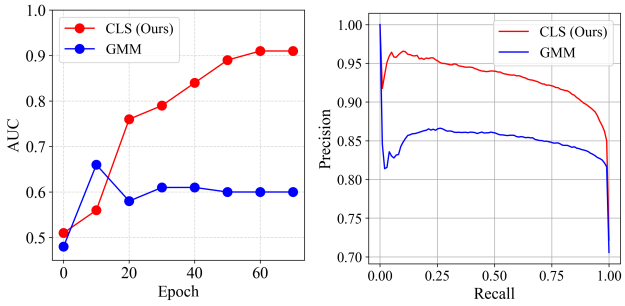


Figure 6: The proposed CKP can be readily integrated with existing LReID and LNL methods to significantly improve their Noisy LReID performance.



(a) Label scoring AUC tendency. (b) Model prediction ROC curve.

Figure 7: Effectiveness of proposed CLS compared to GMM under 30% random noise.

frequently rectifying the label is not necessary and a rectification interval of 5 epochs is enough to guarantee the performance. In practice, we set T_c, T_o, w_l, and e_0 to 0.8, 0.2, 0.1, and 5 respectively.

Combination with other methods. As is shown in Figure 6, when our method is combined with existing methods, 8.6%-17.7% improvement is achieved. CORE† and CORE‡ represent integrating CORE with the anti-forgetting strategy of LSTKC and DKP, respectively. Note that LSTKC and DKP are inter-instance relation-based knowledge distillation methods and LwF is logits-based knowledge distillation method. These results demonstrate the compatibility of our method with different anti-forgetting strategies.

Effectiveness of Cluster-aware Label Scoring. To evaluate the identity confidence estimation capability of our CLS strategy which plays an important role in our CDP and EKF, we experimentally replace CLS with the widely-used Gaussian Mixture Model (GMM) in our approach. Figure 7 (a) illustrates the tendency of label scoring AUC across training epochs. Initially, GMM performs slightly better, but as the model begins to overfit the label noise, its label scoring capacity diminishes after the 10th epoch. In contrast, our CLS consistently improves its AUC performance and surpasses GMM after the 20th epoch. This shows the superiority of our CLS in guiding the algorithms to collect clean data. Furthermore, we visualize the ROC curves of the predicted identities generated by the final models in Figure 7 (b). Thus, it is evident that our CLS effectively guides the model to learn the correct knowledge.

5 Conclusion

In this paper, we initially investigate a practical task named Noisy Lifelong Person Re-Identification (Noisy LReID), which suffers exacerbated correct knowledge catastrophic forgetting and additional erroneous knowledge catastrophic remembering problems. To facilitate research in Noisy LReID, we introduce a benchmark named NLReID. In addition, we propose a novel and effective Continual Knowledge Purification (CKP) framework. To reduce the erroneous knowledge acquisition, an Iterative Label Rectification pipeline, and a Cluster-aware Data Purification module are designed to rectify the noise labels and collect clean data along the training procedure to mitigate the influence of noisy data on new knowledge learning. Besides, to handle the catastrophic remembering and forgetting issues, an Erroneous Knowledge Filtering algorithm is proposed to reduce erroneous old knowledge accumulation and ensure correct knowledge consolidation. Extensive experiments show our method is robust to different kinds of label noise and achieves significant Noisy LReID performance improvement, especially under high-ratio noise compared to existing methods.

Acknowledgments

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References

[1] Ejaz Ahmed, Michael Jones, and Tim K Marks. 2015. An Improved Deep Learning Architecture for Person Re-identification. In *CVPR*. IEEE, 3908–3916.
[2] Home Office Scientific Development Branch. 2006. Imagery library for intelligent detection systems (i-lids). In *2006 IET Conference on Crime and Security*. IET, 445–448.
[3] Yongchun Chen, Min Liu, Xueping Wang, Fei Wang, An-An Liu, and Yaonan Wang. 2023. Refining noisy labels with label reliability perception for person re-identification. *IEEE Transactions on Multimedia* (2023).
[4] Dengpan Fu, Dongdong Chen, Hao Yang, Jianmin Bao, Lu Yuan, Lei Zhang, Houqiang Li, Fang Wen, and Dong Chen. 2022. Large-Scale Pre-training for Person Re-identification with Noisy Labels. In *CVPR*. IEEE, 2476–2486.
[5] Lihua Fu, Yubin Du, Yu Ding, Dan Wang, Hanxu Jiang, and Haitao Zhang. 2022. Domain Adaptive Learning with Multi-Granularity Features for Unsupervised Person Re-identification. *Chinese Journal of Electronics* 31, 1 (2022), 116–128.
[6] Wenhang Ge, Junlong Du, Ancong Wu, Yuqiao Xian, Ke Yan, Feiyue Huang, and Wei-Shi Zheng. 2022. Lifelong Person Re-identification by Pseudo Task Knowledge Preservation. In *AAAI*, Vol. 36. 688–696.
[7] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao. 2021. Knowledge distillation: A survey. *IJCV* 129, 6 (2021), 1789–1819.
[8] Douglas Gray and Hai Tao. 2008. Viewpoint invariant pedestrian recognition with an ensemble of localized features. In *ECCV*. Springer, 262–275.
[9] Guangrui Guo, Zhonghang Wei, and Jinyong Cheng. 2023. Enhancing Continual Noisy Label Learning with Uncertainty-Based Sample Selection and Feature Enhancement. In *PRCV*. Springer, 498–510.
[10] Shuting He, Hao Luo, Pichao Wang, Fan Wang, Hao Li, and Wei Jiang. 2021. TransReID: Transformer-based Object Re-Identification. In *ICCV*. IEEE, 14993–15002.
[11] Martin Hirzer, Csaba Beleznai, Peter M Roth, and Horst Bischof. 2011. Person re-identification by descriptive and discriminative classification. In *Image Analysis*. Springer, 91–102.
[12] Zhizhong Huang, Junping Zhang, and Hongming Shan. 2023. Twin contrastive learning with noisy labels. In *CVPR*. 11661–11670.
[13] Zhipeng Huang, Zhizheng Zhang, Cuiling Lan, Wenjun Zeng, Peng Chu, Quanzeng You, Jiang Wang, Zicheng Liu, and Zheng-jun Zha. 2022. Lifelong Unsupervised Domain Adaptive Person Re-identification with Coordinated Anti-forgetting and Adaptation. In *CVPR*. IEEE, 14288–14297.
[14] Nazmul Karim, Umar Khalid, Ashkan Esmaeili, and Nazanin Rahnavard. 2022. Cnll: A semi-supervised approach for continual noisy label learning. In *CVPR*. 3878–3888.
[15] Prakhar Kaushik, Adam Kortylewski, Alex Gain, and Alan Yuille. 2021. Understanding Catastrophic Forgetting and Remembering in Continual Learning with Optimal Relevance Mapping. In *Fifth Workshop on Meta-Learning at the Conference on Neural Information Processing Systems*.
[16] Chris Dongjoo Kim, Jinseo Jeong, Sangwoo Moon, and Gunhee Kim. 2021. Continual learning on noisy data streams via self-purified replay. In *ICCV*. 537–547.
[17] Xu Lan, Hanxiao Wang, Shaogang Gong, and Xiatian Zhu. 2017. Deep reinforcement learning attention selection for person re-identification. *arXiv preprint arXiv:1707.02785* (2017).
[18] Wei Li and Xiaogang Wang. 2013. Locally aligned feature transforms across views. In *CVPR*. IEEE, 3594–3601.
[19] Wei Li, Rui Zhao, and Xiaogang Wang. 2012. Human Reidentification with Transferred Metric Learning. In *ACCV*. Springer, 31–44.
[20] Wei Li, Rui Zhao, Tong Xiao, and Xiaogang Wang. 2014. DeepReID: Deep Filter Pairing Neural Network for Person Re-identification. In *CVPR*. IEEE, 152–159.
[21] Wei Li, Xiatian Zhu, and Shaogang Gong. 2018. Harmonious Attention Network for Person Re-identification. In *CVPR*. IEEE, 2285–2294.
[22] Yifan Li, Hu Han, Shiguang Shan, and Xilin Chen. 2023. Disc: Learning from noisy labels via dynamic instance-specific selection and correction. In *CVPR*. 24070–24079.
[23] Zhizhong Li and Derek Hoiem. 2017. Learning without Forgetting. *PAMI* 40, 12 (2017), 2935–2947.
[24] Chen Change Loy, Tao Xiang, and Shaogang Gong. 2010. Time-delayed Correlation Analysis for Multi-camera Activity Understanding. *IJCV* 90, 1 (2010), 106–129.
[25] Wonpyo Park, Dongju Kim, Yan Lu, and Minsu Cho. 2019. Relational knowledge distillation. In *CVPR*. IEEE, 3967–3976.
[26] Nan Pu, Wei Chen, Yu Liu, Erwin M Bakker, and Michael S Lew. 2021. Lifelong Person Re-Identification via Adaptive Knowledge Accumulation. In *CVPR*. IEEE, 7897–7906.
[27] Nan Pu, Yu Liu, Wei Chen, Erwin M Bakker, and Michael S Lew. 2022. Meta reconciliation normalization for lifelong person re-identification. In *ACMM MM*. 541–549.
[28] Nan Pu, Zhun Zhong, Nicu Sebe, and Michael S Lew. 2023. A Memorizing and Generalizing Framework for Lifelong Person Re-Identification. *PAMI* (2023).
[29] Ergys Ristani, Francesco Solera, Roger Zou, Rita Cucchiara, and Carlo Tomasi. 2016. Performance measures and a data set for multi-target, multi-camera tracking. In *ECCV*. Springer, 17–35.
[30] Erich Schubert, Jörg Sander, Martin Ester, Hans Peter Kriegel, and Xiaowei Xu. 2017. DBSCAN revisited, revisited: why and how you should (still) use DBSCAN. *ACM Transactions on Database Systems (TODS)* 42, 3 (2017), 1–21.
[31] James Seale Smith, Junjiao Tian, Shaunak Halbe, Yen-Chang Hsu, and Zsolt Kira. 2023. A closer look at rehearsal-free continual learning. In *CVPR*. 2409–2419.
[32] Chunfeng Song, Yan Huang, Wanli Ouyang, and Liang Wang. 2018. Mask-guided contrastive attention model for person re-identification. In *CVPR*. 1179–1188.
[33] Zhicheng Sun and Yadong Mu. 2022. Patch-based Knowledge Distillation for Lifelong Person Re-Identification. In *ACM MM*. 696–707.
[34] Marco Toldo and Mete Ozay. 2022. Bring evanescent representations to life in lifelong class incremental learning. In *CVPR*. IEEE, 16732–16741.
[35] Yuanpeng Tu, Boshen Zhang, Yuxi Li, Liang Liu, Jian Li, Jiangning Zhang, Yabiao Wang, Chengjie Wang, and Cai Rong Zhao. 2023. Learning with noisy labels via self-supervised adversarial noisy masking. In *CVPR*. 16186–16195.
[36] Weitao Wan, Yuanyi Zhong, Tianpeng Li, and Jiansheng Chen. 2018. Rethinking feature distribution for loss functions in image classification. In *CVPR*. 9117–9126.
[37] Zifeng Wang, Zizhao Zhang, Sayna Ebrahimi, Ruoxi Sun, Han Zhang, Chen-Yu Lee, Xiaoqi Ren, Guolong Su, Vincent Perot, Jennifer Dy, et al. 2022. DualPrompt: Complementary Prompting for Rehearsal-free Continual Learning. *arXiv:2204.04799* (2022).
[38] Longhui Wei, Shiliang Zhang, Wen Gao, and Qi Tian. 2018. Person Transfer GAN to Bridge Domain Gap for Person Re-identification. In *CVPR*. IEEE, 79–88.
[39] Guile Wu and Shaogang Gong. 2021. Generalising without Forgetting for Lifelong Person Re-identification. In *AAAI*, Vol. 35. 2889–2897.
[40] Tong Xiao, Shuang Li, Bochao Wang, Liang Lin, and Xiaogang Wang. 2016. End-to-end deep learning for person search. *arXiv:1604.01850* 2, 2 (2016), 4.
[41] Kunlun Xu, Xu Zou, Yuxin Peng, and Jiahuan Zhou. 2024. Distribution-aware Knowledge Prototyping for Non-exemplar Lifelong Person Re-identification. In *CVPR*. 16604–16613.
[42] Kunlun Xu, Xu Zou, and Jiahuan Zhou. 2024. LSTKC: Long Short-Term Knowledge Consolidation for Lifelong Person Re-identification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 16202–16210.
[43] Rui Xu and Donald Wunsch. 2005. Survey of clustering algorithms. *IEEE Transactions on neural networks* 16, 3 (2005), 645–678.
[44] Mouxing Yang, Zhenyu Huang, Peng Hu, Taihao Li, Jiancheng Lv, and Xi Peng. 2022. Learning with twin noisy labels for visible-infrared person re-identification. In *CVPR*. 14308–14317.
[45] Mouxing Yang, Zhenyu Huang, and Xi Peng. 2024. Robust object re-identification with coupled noisy labels. *IJCV* (2024), 1–19.
[46] Zexian Yang, Dayan Wu, Wanqian Zhang, Bo Li, and Weipinng Wang. 2023. Handling label uncertainty for camera incremental person re-identification. In *ACM MM*. 6253–6263.
[47] Mang Ye, He Li, Bo Du, Jianbing Shen, Ling Shao, and Steven CH Hoi. 2021. Collaborative refining for person re-identification with label noise. *TIP* 31 (2021), 379–391.
[48] Mang Ye and Pong C Yuen. 2020. PurifyNet: A robust person re-identification model with noisy labels. *TIFS* 15 (2020), 2655–2666.
[49] Zhaoda Ye, Xiangteng He, and Yuxin Peng. 2022. Unsupervised Cross-Media Hashing Learning via Knowledge Graph. *Chinese Journal of Electronics* 31, 6 (2022), 1081–1091.
[50] Li Yi, Gezheng Xu, Pengcheng Xu, Jiaqi Li, Ruizhi Pu, Charles Ling, A Ian McLeod, and Boyu Wang. 2023. When source-free domain adaptation meets learning with noisy labels. *arXiv preprint arXiv:2301.13381* (2023).
[51] Chunlin Yu, Ye Shi, Zimo Liu, Shenghua Gao, and Jingya Wang. 2023. Lifelong Person Re-Identification via Knowledge Refreshing and Consolidation. In *AAAI*, Vol. 37. 3295–3303.
[52] Tianyuan Yu, Da Li, Yongxin Yang, Timothy M Hospedales, and Tao Xiang. 2019. Robust person re-identification by modelling feature uncertainty. In *ICCV*. IEEE, 552–561.
[53] Won Joon Yun, Jihong Park, and Joongheon Kim. 2023. Quantum multi-agent meta reinforcement learning. In *AAAI*, Vol. 37. 11087–11095.
[54] Ruru Zhang, E Haihong, Meina Song, and Xun Cao. 2024. FSCIL-EACA: Few-Shot Class-Incremental Learning Network Based on Embedding Augmentation and Classifier Adaptation for Image Classification. *Chinese Journal of Electronics* 33, 1 (2024), 139–152.
[55] Haiyu Zhao, Maoqing Tian, Shuyang Sun, Jing Shao, Junjie Yan, Shuai Yi, Xiaogang Wang, and Xiaoou Tang. 2017. Spindle Net: Person Re-identification with Human Body Region Guided Feature Decomposition and Fusion. In *CVPR*. IEEE, 907–915.

[56] Haiyan Zhao, Tianyi Zhou, Guodong Long, Jing Jiang, and Chengqi Zhang. 2023. Does continual learning equally forget all parameters?. In *ICML*. PMLR, 42280–42303.

[57] Rui Zhao, Wanli Ouyang, and Xiaogang Wang. 2013. Unsupervised salience learning for person re-identification. In *CVPR*. 3586–3593.

[58] Liang Zheng, Liyue Shen, Lu Tian, Shengjin Wang, Jingdong Wang, and Qi Tian. 2015. Scalable person re-identification: A benchmark. In *ICCV*. IEEE, 1116–1124.